MILITARY SPECIFICATION

BOILERS, FLUIDIZED BED, STEAM AND HOT WATER, HIGH AND LOW PRESSURE, FIRETUBE, PACKAGED TYPE (2,000,000 TO 50,000,000 BTU/H)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal firetube, multipass fluidized bed combustion (FBC), shop-assembled, packaged, steam boiler (steam generator) and hot water generators equipped with automatic flame safeguard control, supervised manual programming control and automatic combustion control with all appurtenances and auxiliary equipment with multi-fuel firing capabilities.

1.2 Classification.

1.2.1 Boiler types. Boilers shall be of the following types, as specified (see 6.2):

Type I - Low pressure steam generator (15 pounds per square inch gage (psig) operating pressure).
Type II - High pressure steam generator (100, 125, 150, 200, 250, and 300 psig design pressures as specified (see 6.2)).
Type III - Low pressure hot water generators (30, 60, 75, 100, and 160 psig design pressures as specified (see 6.2); temperatures not exceeding 250° Fahrenheit (F)).

1.2.2 Boiler sizes. For the purpose of defining requirements, boilers covered by this specification are classified under the following sizes based on the output in British thermal units per hour (Btu/h). The actual boiler capacity shall be as specified (see 3.8.2, and 6.2):

Size 1 - 2,000,000 to 2,500,000 Btu/h.
Size 2 - 2,500,001 to 5,000,000 Btu/h.
Size 3 - 5,000,001 to 10,000,000 Btu/h.
Size 4 - 10,000,001 to 20,000,000 Btu/h.
Size 5 - 20,000,001 to 30,000,000 Btu/h.
Size 6 - 30,000,001 to 40,000,001 Btu/h.
Size 7 - 40,000,001 to 50,000,001 Btu/h.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Burner types and control systems. Burner furnished on the boilers shall be of the following types and control systems, as specified (see 6.2):

Type CLO - Combination coal-fired and light-oil-fired (grade 2 or 4, as specified).
Type CHO - Combination coal-fired and heavy-oil-fired (grade 5 or 6, as specified).
Type CNG - Combination coal-fired and natural-gas-fired.
Type CGL - Combination coal-fired, gas-fired, and light-oil-fired (type of gas and grade of oil as specified).
Type CGH - Combination coal-fired, gas-fired, and heavy-oil-fired (type of gas and grade of oil as specified).

Combustion control system:

Parallel (positioning with air/fuel ratio control) (see 3.11.6.1).
Parallel (positioning with air/fuel ratio control and oxygen compensation) (see 3.11.6.2).

2. APPLICABLE DOCUMENTS.

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball Annular (General Purpose).
GG-G-76 - Gages, Pressure and Vacuum Dial Indicating (for Air, Steam Oil, Water, Ammonia, and Chloro-floro Hydro Carbon Gasses).
TT-P-28 - Paint, Aluminum, Heat Resisting (1,200°F).
VV-F-800 - Fuel Oil, Diesel.
ZZ-B-225 - Belt, V, Industrial (Multiple Drive).

MILITARY

MIL-V-173 - Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-F-859 - Fuel Oil, Burner.
MIL-B-3180 - Boiler and Related Equipment, Packaging of.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-T-13867 - Treatment, Moisture and Fungus Resistant for Fire Control Electrical and Electronic Instruments and Equipment.
MIL-P-17749 - Pumping Units, Condensate, Heating, and Pumping Units, Vacuum, Heating.
MIL-R-18115 - Regulator, Boiler Feed Water, Automatic.
MIL-V-18406 - Valve, Blowoff, Boiler.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

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STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF LABOR (DoL)
OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

Occupational Safety and Health Standards.

(Application for copies of DoL and OSHA should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

AIR MOVING AND CONDITIONING ASSOCIATION (AMCA)

Standard 210D - Test Code for Air Moving Devices.

(Application for copies should be addressed to the Air Moving and Conditioning Association, 30 W. University Drive, Arlington Heights, IL 60004.)

AMERICAN BOILER MANUFACTURERS ASSOCIATION (ABMA)

Packaged Firetube Boiler Ratings.
Lexicon Steam Generating Equipment.

(Application for copies should be addressed to the American Boiler Manufacturers Association, Suite 700, 1500 Wilson Blvd., Arlington, VA 22209.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B2.1 - Pipe Threads (Except Dryseal)
B15.1 - Mechanical Power Transmission Apparatus, Safety Standard for.
B31.1 - Power Piping.
Z83.3 - Gas Utilization Equipment in Large Boilers.
Z223.1 - National Gas Fuel Code.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF HEATING REFRIGERATING AND AIR - CONDITIONING ENGINEERS INC (ASHRAE)

No. 85 - Automatic Control Terminology for Heating, Air-Conditioning, and Refrigeration Equipment.

(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air-Conditioning Engineers Inc., 345 East 47th Street, New York, N.Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

CSD-1 - Controls and Safety Devices for Automatically Fired Boilers.
Boiler and Pressure Vessel Code.
Section I - Power Boilers.
Section IV - Heating Boilers.
Section VIII - Pressure Vessels.
ASME Performance Test Code.
PTC 4.1 - Steam Generating Units.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A588 - High Strength Low-Alloy Structural Steel with 50,000 psi Minimum Yield Point to 4 Inch Thick, Specification for.
C64 - Refractories for Incinerators and Boilers, Specification for.
D396 - Fuel oils.
D1757 - Sulfur in Coal Ash.
D2156 - Smoke Density in the Flue Gases from Burning Distillate Fuels, Test Method for.
E230 - Temperature, Electromotive Force (EMF) Table for Thermocouples.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

CODE OF FEDERAL REGULATIONS (CFR)

Title 40, Part 60.344 - Visible Emissions - Method 9.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852.)

FACTORY MUTUAL SYSTEM (FM)

Factory Mutual Approval Guide.

(Application for copies should be addressed to the Factory Mutual System, 1151 Boston-Providence Turnpike, Norwood, MA 02062.)

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INSTRUMENT SOCIETY OF AMERICA (ISA)

S18.1 - Annunciator Sequences and Specifications.

(Application for copies should be addressed to the Instrument Society of America, 67 Alexander Drive, P.O. Box 12277, Research Triangle Park, NC 27709.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ICS-1 - General Standards for Industrial Controls and Systems.
ICS-2 - Standards for Industrial Control Devices, Controllers, and Assemblies.
ICS-6 - Enclosures for Industrial Control Systems.
MG-1 - Standards for Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electric Code.
No. 85B - Furnace Explosions in Natural Gas-Fired Multiple Burner Boiler-Furnaces.
No. 85D - Fuel Oil-Fired Multiple Burner Boiler-Furnaces.
No. 85E - Pulverized Coal-Fired Multiple Burner Boiler-Furnaces.
No. 86A - Ovens and Furnaces.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

RUBBER MANUFACTURERS ASSOCIATION

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

UNDERWRITERS' LABORATORIES, INC. (UL)

UL353 - Limit Controls.
UL726 - Standard for Oil-Fired Boilers.
UL795 - Commercial-Industrial Gas-Heating Equipment.
Gas and Oil Equipment List for Gas, Gas-Oil, and Oil Burners.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The steam boilers (steam generators) and hot water generators shall be horizontal firetube, multipass, fluidized bed combustion type, packaged units mounted on a skid-type structural steel base. Each

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generator shall be equipped with a complete burner and fuel system; a completely water-cooled furnace; forced and induced draft fans; an automatic electronic control system complete with combustion and flame safeguard controls, firing sequence programmer, safety interlocks, limit controls, and central control panel; and such trim and appurtenances as are peculiar to steam units and water units as specified herein. All units shall be factory-wired and assembled except for such readily installed appurtenances as safety valves, water columns, pressure gages, oil heater piping, draft fans, and fuel systems. Such items and assemblies may be removed prior to shipment to prevent damage, reduce cubage, or meet shipping dimension restrictions, or installation requirements. In other respects, the units shall be complete and ready for operation when connected to water, fuel, electrical supplies, and draft connections.

3.2 First article. When specified (see 6.2), the contractor shall furnish a steam generator or hot water generator for first article inspection and approval (see 6.5).

3.3 Standard commercial product. Each boiler of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the boiler being furnished. A standard commercial product is a product which has been sold or is currently being offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Standards compliance. Steam generators and hot water generators, exclusive of any supplementary control devices, safety interlocks, or limit controls required under this specification, shall meet the requirements of the following standards:

(a) Oil-fired units - UL726.
(b) Gas-fired units - UL795.
(c) All units - ASME Boiler and Pressure Vessel Code.

Any supplementary controls furnished in compliance with this specification shall be UL listed.

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3.5.1 Certification. Prior to approval of the sample generator furnished for first article inspection, if such inspection is specified, or prior to the first shipment, if first article inspection is not required, the contractor shall submit evidence satisfactory to the contracting officer or his authorized representative that the generators to be furnished under this specification meet the UL standards cited in 3.5. Acceptable evidence of meeting the specified requirements herein shall be certificate of compliance (see 6.3), executed by an authorized official of the contractor, stating that the items have been tested and that the units meet the requirements of the specified standards in 3.5, including specified methods of testing. Acceptable evidence of meeting the ASME Boiler and Pressure Vessel Code may be the applicable code symbol stamp or a certified test report (see 6.3) from a recognized independent laboratory acceptable to the Government, indicating that the generator meets the requirements of the applicable section of the ASME Boiler and Pressure Vessel Code. Acceptable evidence of meeting the fuel and combustion controls, combustion safeguard controls, and safety shutoff valves will be the official UL listing mark prescribed in the UL Gas and Oil Equipment List for Gas, Gas-Oil, and Oil Burners, FM approval mark prescribed in the Approval Guide of the Factory Mutual System, or a certified test report (see 6.3), from a recognized independent laboratory acceptable to the Government, indicating that the components meet the approval requirements of UL or FM.

3.6 Interchangeability. All generators of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts assemblies, accessories, and spare parts.

3.7 Design. The high pressure steam generators shall be designed in accordance with the requirements of Section I of the ASME Boiler and Pressure Vessel Code applicable to firetube boilers. Low pressure steam generators and hot water generators shall be designed in accordance with section IV of the ASME Code.

3.7.1 Design pressure. The design pressures specified herein shall be the maximum allowable working pressure for the generators as determined by employing the stress values, design rules, and dimensions specified in the applicable section of the ASME Code. The design pressure for low pressure steam generators shall be 30 psig. The design pressures for low pressure hot water generators and high pressure steam generators shall be as specified in 1.2.1.

3.7.2 Design for thermal shock. Hot water generators shall be designed to withstand water temperature differentials associated with single zone and multi-zone hot water heating systems without danger of damage from stresses induced by thermal shock. Hot water generators designed for natural circulation shall be equipped with supply and return connections sized and located in accordance with the manufacturer's standard practice to provide sufficient natural thermal circulation for minimizing the effects of thermal shock.

3.8 Performance. Unless otherwise specified (see 6.2), the performance requirements specified herein shall be based on the following criteria.

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A Coal analysis (percent by weight unless noted)

	Percent
Moisture	17.5 max
Volatile matter	42 max
Ash	10 max
Sulphur	4 min
Fixed Carbon	remainder
Gross Calorific value (Btu 1 lb)	12700 min
Ash melting temperature	2100°F min
Grindability Index	59 min
Free-swelling index	3 to 5

B Limestone or dolomite (percent by weight unless noted)

	Percent
Silica (SiO_2)	5.28
Aluminum oxide (Al_2O_3)	0.76
Iron oxide (Fe_2O_3)	0.76
Calcium carbonate ($CaCO_3$)	58.86
Magnesium carbonate ($MgCO_3$)	36.97
Sulfur trioxide (SO_3)	.34

Size: 1/16 to 1/8 inch.

C Fuel oil (percent by weight, as fired).

Ultimate analysis:

H_2	10.33
C	87.87
O_2	0.25
N_2	0.14
S	1.16
ASH	0.25
Btu/lb	18,400

D Natural gas (percent by volume, as fired).

Ultimate analysis:

Methane (CH_4)	90.00
Ethane (C_2H_6)	5.00
Nitrogen	5.00
ASH	NIL
Moisture	0.01
Btu/cubic foot (cu ft)	1002
Btu/lb	21,824

The gas supply pressure downstream of the service regulator to be available for use at the inlet to the gas burner pressure regulator shall be as specified in the contract (see 6.2). Fuel oils, No.'s 2 through 6 shall conform to the requirements of ASTM D396. Navy special fuel oil (NSFO) shall conform to the requirements of MIL-F-859. Diesel shall conform to VV-F-800 and JP-5 shall conform to the requirements of MIL-T-5624.

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3.8.1 Boiler output conditions. The operating pressure for steam generators shall be as specified (see 6.2) and shall be the pressure at which the boiler operating controls shall be set by the manufacturer prior to shipment. Unless otherwise specified (see 6.2), the operating temperature for type III hot water generators at which the temperature-sensing operating controls shall be set by the manufacturer prior to shipment shall be 200°F. For steam generators the moisture content of saturated steam, produced from generator water having a total dissolved solids concentration of 2200 parts per million and a total alkalinity not in excess of 440 parts per million, shall not exceed one percent during a change within any three-minute period from 50 percent to 100 percent of rated capacity as well as during continuous operation at full rated capacity.

3.8.2 Capacity. The maximum continuous thermal output of steam generators and hot water generators, expressed in Btu/h at the outlet connection, shall be as specified (see 1.2.2 and 6.2) and shall be based on the conditions as defined in 3.7, 3.8, and as may be amplified or modified under provisions of the contract documents. If the procurement documentation should specify capacity for steam generators in terms of boiler horsepower (hp), or pounds of steam, the thermal output shall be computed on the basis of 33,475 Btu/h per boiler hp, or 970 Btu/h per pound of steam, at an operating pressure of 14.7 psi absolute with a feedwater temperature of 212°F (from and at 212°F). In either case, the specified capacity shall be attained within the limits for heat release and heat transfer specified herein.

3.8.3 Efficiency.

3.8.3.1 Combustion efficiency. In accordance with the ABMA Standard Test Procedures for Packaged Firetube Boilers, the combustion efficiency, carbon dioxide, carbon monoxide, and oxygen for fuels fired herein shall be as specified in table I and shall be not less than the average of the three rated loads of table I. When fired with fuels herein, the generator shall not exceed a smoke density reading in accordance with ASTM D2156 (Shell-Bacharach) of the values listed in table I, except for short periods not to exceed 3 minutes. When specified (see 6.2), the generators shall be designed to burn the fuels fired herein, so that the particulate, nitrogen oxide, and sulfur dioxide emissions as specified (see 6.1(k) and 6.2) shall not exceed the limits when determined in accordance with 4.5.4. Both stable firing and combustion efficiency shall be maintained over the entire firing range required by the boiler turndown (see 3.12) as specified herein.

3.8.3.2 Fuel-to-steam efficiency. In accordance with the ABMA Standard Test Procedure for Packaged Firetube Boilers or ASME PTC 4.1, the fuel-to-steam efficiency for type I generators shall be 80 percent and for type II generators shall be 78 percent when fired at maximum rated output established in accordance with the provisions of 3.8, 3.8.1, and 3.8.2.

3.8.3.3 Fuel-to-hot water efficiency. In accordance with the ABMA Standard Test Procedure for Packaged Firetube Boilers or ASME PTC 4.1, the fuel-to-hot water efficiency for type III generators shall be 80 percent when fired at maximum rated output established in accordance with the provisions of 3.8, 3.8.1, 3.8.2.

3.8.4 Thermal input and output. When the units are operating at the maximum rated output established in accordance with the provisions of 3.8, 3.8.1, and 3.8.2, the thermal input to the furnace shall be such that none of the following rates specified herein will be exceeded:

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(a) Heat release rate per cubic foot (cu ft) of furnace volume per hour.
(b) Heat release rate per square foot (sq ft) of effective radiant heating surface (ERHS) per hour.
(c) Heat transfer rate per sq ft of total boiler heating surface per hour based on thermal output.

3.8.4.1 Volume heat input rate. The volume heat input rate (see 6.4), shall not exceed 80,000 Btu/h/cu ft of the furnace volume for coal firing. The above values are based on boilers with a total ERHS equal to 90 percent of the total wall furnace area being water-cooled.

3.8.4.2 Surface heat input rate. The maximum heat input rate of effective radiant heating surface of the furnace shall not exceed 60,000 Btu/h/sq ft (see 6.4).

3.8.4.3 Heat transfer rate. The heat transfer rate shall be determined on the basis of the furnace input times the fuel-to-steam or fuel-to-hot water efficiency divided by the total generator heating surface (see 6.4) and shall not exceed 8,400 Btu/h/sq ft.

3.8.5 Pressure and temperature regulation. The combustion control systems shall maintain the steam pressure or water temperature, as applicable, at the outlet connection within ±7 percent expressed as a percent of the set point values of the operating control.

3.8.6 Exit temperature at stack. The exit temperature at the stack of the generator or economizer shall not exceed 125°F above the dew point temperature of the steam or above the exit temperature of the hot water.

3.8.7 Sulfur retention in the ash. When firing coal the generator shall retain 90 percent of the sulfur in the ash of the coal being fired as determined by ASTM D1757. Acceptable evidence of meeting the sulfur retention requirements shall be a certified test report from a recognized independent laboratory acceptable to the Government (see 6.3).

3.9 Construction. The steam generators and hot water generators shall be constructed in accordance with the applicable section of the ASME Boiler and Pressure Vessel Code for the design pressure specified herein.

3.9.1 Shell. The generator shell shall be a horizontal, cylindrical, steel pressure vessel of sufficient size to meet the performance requirements of this specification. The generator shell shall be equipped with all necessary connections including outlet nozzles, return connections, and connections for pressure relief valves, water level controls, and other required trim. Manholes and handholes shall be provided in accordance with the ASME Boiler and Pressure Vessel Code.

3.9.2 Furnace. The furnace shall be a fusion-welded chamber welded to the shell. The furnace sidewalls and roof shall be 100 percent water-cooled surface. Heat distribution shall be uniform under all firing conditions with no flame impingement (see 6.4). The furnace shall be so constructed that it is completely surrounded by water at all times and shall not interfere with the natural circulation of water within the shell and to assure the reduction of the furnace exit gas temperatures to a maximum of 1,600°F prior to entry into the firetube convection zone. The furnace shall be positioned at a sufficient distance from the shell so that the shell will not be subjected to unequal thermal stresses during the start up or any other condition of firing or operation.

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3.9.3 Firetubes and watertubes. Firetubes and watertubes shall be of steel and shall be of the seamless type. The tubes shall be fitted into accurately sized holes in the tube sheets and tightly rolled in place. The alinement shall be such as to prevent noticeable deformation or undue stress when the generator is put in service. The tube and tube sheet assembly shall be water- and gas-tight. The tubes shall be so arranged as not to interfere with the natural circulation of water in the shell or to prevent the cleaning and flushing of all watersides. All firetubes shall be readily removable from one end of the generator, either the front or the rear. Spinners, turbulators, and other such devices shall not be permitted in the firetubes.

3.9.4 Refractories. Refractories, when provided, shall be capable of withstanding the temperature existing under maximum load conditions in accordance with ASTM C-64. The refractories shall be formed or cast in sections of such size as to be easily replaceable through existing openings. The refractory sections shall be designed and secured in position in such a manner as to withstand the vibration and shocks occurring during shipment. Refractories used for the combustion chamber shall have a pyrometric cone equivalent (PCE) of not less than 33. Refractory shall be provided for all doors and end covers exposed to temperatures of 600°F and over.

3.9.5 Door, access ports, and observation ports. Flue doors and cleanout openings shall be furnished to provide full access to all fireside surfaces for inspection, cleaning, and repair, and shall have sufficient insulation on the fireside to prevent the heat from damaging the operating mechanism or otherwise impairing the operation of the door or device. Doors weighing over 50 lb shall be hinged or davited. When specified (see 6.2), a spring-loaded relief door or other approved explosion relief device shall be furnished to relieve excessive pressures in the combustion spaces. Sealed heat resisting observation ports shall be furnished to provide full observation of maintenance of combustion conditions including lancing and fluidized bed coal burners. All doors and observation ports shall be gas-tight under all conditions of firing. Flue doors shall be readily removable without impairment to the refractory, when provided, and without requiring the use of additional refractory binders, cements, or sealing materials.

3.9.6 Insulation. All exterior surfaces of the unit, except the doors, end covers, handholes, manholes, and vents shall be covered where practical with at least two inches of fibrous glass, mineral wool, thermal block, or equivalent insulation having a heat transfer coefficient at a mean temperature of 300°F not exceeding 0.45 Btu/h per square foot per inch thickness per degrees F temperature difference. The insulation shall be so formed and secured in place as to prevent sagging or displacement during shipment and operation.

3.9.7 Outer casing. The insulation shall be covered with a sheet steel jacket not less than .0289 thick (U.S. revised standard gage No. 22) thick. The average surface temperature of the casing shall not exceed 130°F in an ambient temperature of 70°F and a surface wind velocity of two feet per second while the boiler is operating at full capacity.

3.9.8 Carbon reinjection. A suitable means for the reinjection of the carbon from the combustion gases to the furnace shall be provided. The sides of the carbon collection chamber shall be at such an angle as to prevent bridging over of the carbon particles. The 125 psig air for carbon reinjection shall be made available by others or the Government as specified (see 6.2). The manufacturer of the generator shall specify the required volume of standard cubic feet per minute (cfm) of air (see 6.4).

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3.9.9 Mechanical dust collector. The collector shall be of the multicone-type and shall include cylindrical body and tapered cone arrangements so designed as to produce effective separation of the dust particles from the air stream in which it is carried with a minimum of dust carryover into the atmosphere. The combustion gases from the boiler shall be vented to the inlet of the collector. An airtight dust receptacle shall be provided as an integral part of the generator. Suitable means for emptying the collected dust particles from the receptacle shall be provided. The collector shell and all sheet metal parts shall be constructed of steel. All joints and seams shall be made airtight, and lap joints shall be in the direction favoring the flow of air. The exhaust vent of the collector shall be connected to the stack of the generator. The collector shall allow not more than 2.0 percent by weight of conveyed material, 50 microns and larger entering the dust collector unit to escape through the stack exhaust. The resistance of the unit shall not exceed 8.0 velocity heads when the velocity is measured at the collector unit inlet.

3.9.10 Ash disposal. The ash from the collector shall be collected into a chamber for manual or conveyer removal, as specified (see 6.2). A connection from the ash collection chamber to the furnace shall be provided for reintroduction of the fly ash to the furnace.

3.9.11 Skid base. The generator shall be mounted on steel saddles or equivalent steel supports fastened securely to a structural steel base designed and constructed to make a complete self-contained unit that will require only a flat surface, capable of supporting the weight of the generating unit when filled with water, for a foundation.

3.9.12 Lifting attachments. Suitable lifting attachments and tiedown devices shall be furnished in accordance with the manufacturer's standard practice except as specified herein. The attachments and tiedown devices shall have eyes not less than 3 inches in inside diameter, and plates to withstand any handling conditions encountered. The tiedown devices shall be located at each corner of the generator. Adequate tiedown devices may also be used for lifting purposes. Each attachment or tiedown device shall be of sufficient capacity to carry the total weight.

3.10 Trim. Each steam generator and hot water generator shall be fully equipped with the trim required under the applicable section of the ASME Boiler and Pressure Vessel Code plus the additional appurtenances specified herein. When disassembly is required for shipping clearance and prevention of damage during shipment, such trim and appurtenances may be field installed. The generator steam and hot water trim, valves, connecting piping and fittings shall be suitable for the design pressure as specified herein. The trim and appurtenances shall include but not necessary be limited to the following:

(a) A six-inch minimum pressure gage with a graduated range that will result in the normal pressure indication being within middle range of the scale. Pressure gages for high pressure steam units shall include a syphon, gage cock, and test connection, and shall be mounted on generator shell.
(b) An ASME rated safety valve(s) for steam units as required by ASME Boiler and Pressure Vessel Code.
(c) An ASME rated safety-relief valve for hot water units as required by ASME Boiler and Pressure Vessel Code.

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(d) Low pressure and high pressure steam generators shall be equipped with a water column complete with sight glass, try cock, gage cocks and blowdown valves. The try cock and valves shall be equipped with levers, chain lengths, and handles for floor level operation. The water column shall be equipped with means to sound low and high water level horn alarm. Drain line piping for water column and gage glass shall terminate not more than five feet above operating floor with a manual valve.

(e) Both low pressure steam units and hot water units shall be equipped with a generator drain valve sized in accordance with the ASME Code.

(f) High pressure steam units shall be equipped with one slow-opening and one quick-opening blowoff valve or a unit-tandem blowoff valve. Blowoff valves shall meet the requirements of MIL-V-18406.

(g) High and low pressure steam units shall be equipped with a continuous blowdown stop valve of the angle type with screwed connections having stainless steel disc and seats.

(h) Continuous blowdown control valve with screwed connections, internal piping from point of highest solids concentration of dissolved solids, and stainless steel disc and seats. The valve shall incorporate means to enable valve opening to be set as desired.

(i) Low water level fuel cutout bypass momentary push button station. The push button station is to be located adjacent to the water column and auxiliary low water cutout drain line piping with manual valves. The push button station shall provide for uninterrupted generator operation for all fuels during either water column or low water level fuel cutout chamber blowdown.

(j) Air vent valve with screwed connection, and stainless steel disc and seats to vent entrapped air from the generator.

(k) Main steam "non return" valve of the angle "Y" pattern, stop check type with flanged connections. Valve shall be sized on the basis of the maximum steam generating capacity, taking into consideration flow characteristics and pressure drop at the varying flow conditions of the operating range when testing the generator and shall not be less than the generator steam outlet. Valves operating over 15 psig shall be cast steel.

(l) A chemical feed connection valve with connected integral distributing pipe shall be provided.

(m) Provide steam main with a calorimeter connection.

(n) The generator fitted with a thermometer well at the feedwater connection.

3.10.1 Feedwater regulator. Regulators shall be mounted on or cross-connected to the water column. The feedwater piping for high pressure steam units shall include a stop-check valve or a separate check valve and gate valve. The regulator shall include as an intergal part the low water cutout of table II. The feedwater regulators shall conform to UL353.

3.10.1.1 Primary. Each steam generator shall be equipped with a primary feedwater regulator of the type III (electrode-actuated) or type II (float-actuated type) in accordance with MIL-R-18115.

3.10.1.2 Secondary. Each steam generator shall be equipped with a secondary feedwater regulator of the type II (float-actuated type) or type III (electrode-actuated quarded ring type) confoming to MIL-R-18115.

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3.10.1.3 Hot water generator. Each hot water generator shall be equipped with one feedwater regulator of the type II (float-actuated type) or type III (electrode-actuated guarded ring type) conforming to MIL-R-18115 for low water cutout.

3.11 Control systems. Each steam generator and hot water generator shall be equipped with the following control systems:

(a) Programming control systems: In conjunction with the electronic flame safeguard control, safety interlocks and limit controls, the programming control systems shall govern the pre-firing, light-off, and shutdown sequences.
(b) Combustion control system: This system shall automatically regulate the firing cycle in accordance with the load demand.

The basic programming and flame safeguard controls shall be furnished as an integrated system by the boiler manufacturer and shall be designed to function in combination with other system components to insure the functional integrity of the total system. Maximum use of common components for the combustion control system and generator limit interlocks shall be made to control oil, gas, and coal firing. Means shall be provided to start the furnace on natural gas or propane to bring the temperature of the bed up to a set point temperature which will assure combustion of unburned fuel and then switch over to the fuel to be fired. When specified (see 6.2), the furnace shall be started on natural gas or propane, bringing the bed to a minimum set point temperature where oil can be ignited and switching over to oil to bring the bed temperature up to the maximum set point temperature to assure combustion of coal. The control system shall be so arranged that in addition, oil can be burned simultaneously with coal at any percentage necessary or desired up to rated capacity of the generator.

3.11.1 Programming control. The programming control systems shall provide the timed operational sequences as functionally specified herein. Status of start-up the sequence of the operating events shall be identified by the colored indicating lights of Table IV. The programming control system for firing coal, gas, and oil shall be supervised manual, (see 6.4) with gas electric spark-igniter. The specified steps shall perform when initiated manually by the operator and the specified interlock functions shall be automatically satisfied and indicated to the operator by position and by the indicating light. The sequence shall conform to the applicable requirements of NFPA No. 85B, No. 85D, and No. 85E. All manual controls shall be interlocked in such a manner to prevent firing out of sequence. A schematic logic diagram of the programming control system and bed temperature monitoring system shall be furnished (see 6.3). The system's functional requirements are as follows:

(a) The sequencing shall insure that operating events occur in proper order.
(b) Only properly prepared fuel shall be admitted to the fluidized bed burners when there is sufficient ignition energy and correct air flow to ignite the fuel as it enters the furnace, and to burn it continuously and as completely as possible within the confines of the furnace.
(c) For start up conditions, means shall be provided to maintain a constant volumetric total air flow (not less than 25 percent) while fuel is controlled to satisfy start-up rates and within the established limits to be compatible with air flow.

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(d) The sequences shall be followed regardless of whether certain functions are accomplished by interlocks or automatic controls or the unit is operated manually. Different arrangements are permissible if they provide equivalent protection and meet the intent of the required operating sequences.

(e) The starting sequence shall provide a continuous air flow through the unit throughout the starting-up and initial load carrying period of operation at the rate which existed during the purge operation until such time as more fuel and air are required for load in order to insure an air-rich furnace atmosphere and thus establish minimum velocities through the furnace to prevent hazardous accumulations of unburned fuel.

(f) The igniters shall ignite the primary bed first and shall be left on until all beds reach a predetermined set point ignition temperature (see 6.4) to assure combustion of unburned fuel as determined and verified by actual experience with the unit and then shall shut off automatically. If the igniters for operating feeders have been turned off, then those operating feeders shall have satisfactory ignition before any igniter is restarted.

(g) If the temperature of the primary bed drops lower than the set point temperature in (f) above, the igniter shall automatically come back on.

(h) Positive interlocks shall prevent beds from being committed or removed in a random pattern but in a proper sequence defined by operating instructions and verified by actual experience with the unit to minimize laning or stratification of fuel and combustion products.

(i) Beds shall be placed in service as required for load, with the fuel flows and individual register or damper settings which will insure light-off with a wide margin for safety.

(j) During start-up, the total furnace throughout shall not be reduced below the purge flow rate to minimize the possibility of a fuel-rich furnace with the probability of an explosion should some malfunction occur such as a fuel control failure or a flameout.

(k) The required fuel flow shall be maintained. The feeder loading fuel flow and primary air may be used to control fuel flow.

(l) A minimum stop, either mechanical or by control setting, shall be provided so that the primary air through the feeder or bed cannot be reduced below a minimum flow that is found by test to prevent an unstable flame. The minimum setting shall not prevent isolation of the feeder or bed.

(m) The igniters in all of the coal beds served by a coal feeder shall be in service and proven by suitable flame safeguard, whenever any coal bed served by the coal feeder is placed in service.

(n) The furnace shall not be permitted to be fired intermittently but a start-up procedure for continuous firing shall be provided.

(o) The first burner shall be permitted to operate or light-off with only the primary bed served by the main coal feeder, provided a positive means to prevent fuel leakage into idle coal feeders and bed be accomplished with positive interlocks until safe ignition temperature is reached for the specific fuel.

(p) When a fuel bed input is sufficiently high to maintain stable flame and reaches a safe ignition temperature for the specific fuel, the hot bed material may be utilized to ignite each subsequent bed as these beds reach the predetermined set temperature.

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(q) Master fuel trips shall operate in a manner to stop all fuel flow including fuel to igniters into the furnace within a period that will not permit a dangerous accumulation of fuel in the furnace. A master fuel trip shall not initiate a fan trip.

(1) Mandatory automatic master fuel trips:

(A) Loss of induced or active forced draft fans.
(B) Furnace overpressure from tube rupture or other cause.
(C) All fuel inputs below minimum established by the manufacturer.
(D) Loss of all flame.
(E) Partial loss of flame sufficient to introduce a hazardous accumulation of unburned fuel.
(F) The shutdown of safety interlocks or limit switches required in 3.11.3.
(G) Open circuit or short circuit thermocouple of the combustion bed (thermocouple break protection).

(2) Mandatory master fuel trips with alarms and silencing switch - not necessarily automatic:

(A) Failure of the first feeder or bed to operate successfully under the conditions of (i) above.
(B) Loss of energy supply for combustion control, burner control, or interlock systems.
(C) Excess furnace suction.

3.11.2 Flame-failure control system. Each igniter shall be equipped with a standard (nonproprietary) flame-failure control system (see 6.4). The flame-failure control system shall have a repetitive self-checking circuit and shall consist of a flame-sensing element properly connected to the programming control system for shutdown of the burner in the event of ignition failure or flame failure during the firing cycle. Trial for ignition of the ignitor shall be limited to 10 seconds. The igniter and burner shall not be allowed to recycle after ignition failure or main flame failure without manual reset. The flame-failure reaction time, including allowable factory tolerances shall be from 2 to 4 seconds for all burners at rated control voltage in an ambient temperature of 70°F. Flame-sensing elements for igniter flames shall detect only flames of sufficient intensity to safely light the main burner. Flame-sensing elements shall be so connected as to prevent startup, if flame is present or if the contacts are stuck in the flame sensing position. The flame-failure control circuit shall be so designed that the failure of any component shall cause a safety-shutdown, prevent restart of the burner after shutdown, and shall incorporate automatic start-check at the beginning of each startup cycle. Flame-sensing elements shall consist of one type, or a combination of the following types within the limitations indicated:

(a) Ultraviolet cells may be used for coal, gas, or oil flames, but shall not be used for igniter proving unless, (1) the spark is shut off during the last part of the proving period, or (2) the scanner is sighted so as to exclude spark detection, or (3) the scanner is equipped with a sight-tube to exclude spark detection (see 4.5.2.3).

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(b) Infrared (lead sulfide) cells may be used for coal, gas, or oil flame but shall not be used in locations where the cell could be actuated by residual radiation emanating from hot refractory after flame extingishment.
(c) Flame rods shall be used only for gas flames, shall be of the rectification type, and shall be used only if the required length of rod does not exceed 12 inches.

When multiple burners are installed the flame sensor elements shall not detect adjacent burners at any load condition. Either sensor location or adjustable sensory controls may be used to accomplish this discrimination.

3.11.2.1 Repetitive self-checking circuit. The repetitive self-checking circuit shall automatically check all components of the flame-failure system including the flame detector (unless the detector is fail-safe such as flame rods and lead sulfide cells) for flame simulating failures at intervals not more than 6 seconds during the entire period of burner operation. The occurrence of a flame simulating failure anywhere in the flame-failure system or the cessation of the cycle operation of the self-checking circuit shall cause a safety shutdown and lockout.

3.11.3 Safety interlock and limit switches. Each fuel system for the type of service (steam or hot water) as specified in 1.2.3 shall be equipped with the applicable safety interlocks and limit switches as indicated in Table II. The safety interlocks and limit switches shall sense the limit or off-limit conditions indicated, cause a nonrecycling safety shutdown or recycling shutdown (see 6.4), of the generator in the specific cases annotated in Table II, and determine the position of the sensing element switch without disassembly of any part of the device. All nonrecycling shutdowns (see 6.4), (1) shall require a manual reset of the interlock, limit switch, or primary flame safeguard control, (2) shall be signaled by a red indicating light and an audible alarm. The safety interlocks and limit switches shall conform to UL353.

3.11.4 Thermocouples. Each bed of the generator shall be equipped with a thermocouple in a thermocouple protection tube. The thermocouples shall extend into the primary combustion bed sufficient distance to sense the average set operating temperature of the bed. The thermocouple protection tube and the thermocouple insulation material shall be suitable for operation in the 1,600°F oxidizing combustion bed for coal, oil, and gas specified herein. The thermocouple's material shall be Type K of ASTM E230 with a time-constant of not longer then 2 seconds as calculated for air at room temperature and atmospheric pressure with velocity of 65 feet per second. The thermocouple's limit of error reference junction at 32°F shall not exceed 0.75 percent. The thermocouple's connection to the generator shall have tapered threads in accordance with ANSI B2.1. The thermocouples shall terminate in a NEMA Type 4 junction box of cast iron and so located that termination does not exceed 105°C.

3.11.5 Bed temperature indicator. Temperature indicator shall be provided on the control panel to indicate minimum and maximum primary bed temperature. The indicator shall be so connected to the igniter controls so that when the primary bed is placed into operation all the igniters shall be lighted and shall extinguish the igniters only when the primary bed is at operating temperature. The indicator shall be also connected so that when the temperature of the primary bed drops below the minimum set temperature all the igniters shall be automatically restarted and sound an alarm and light a red

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indicating light. The indicator shall be compatible for the thermocouple of 3.11.4, provide thermocouple break protection, and cold junction compensation. The indicator accuracy shall be plus or minus 0.5 percent of span. The indicator shall have acknowledge and reset buttons and be enclosed in a dust tight case suitable for flush mounting. The indicator's electrical input shall not exceed 120V, 60Hz, 2 wire with separate equipment ground rated not less than 5 amperes (A). When specified (see 6.2), the indicator shall record bed temperature on a strip or circular chart as specified (see 6.2).

3.11.6 Combustion controls. Combustion controls specified in 1.2.3, shall automatically regulate the firing rate in accordance with the load demand and shall be capable of meeting the burner turndown (see 6.4) specified in 3.12. Control systems shall maintain air-fuel mixtures within the limits required for stable and efficient combustion throughout the specified firing range and, as applicable, during changes in firing rates. The combustion control systems shall maintain the steam pressure or water temperature, as applicable, at the outlet connection as specified in 3.8.5. The tolerance limits shall be expressed as a percent of the set point values of the operating control. The specified tolerance shall apply to any load which, within a one minute period, swings from a steady state condition to an increase (or decrease) in load equal to a maximum of 10 percent of the full rated capacity of the burner. Regulation tolerances shall apply to any steady-state condition within the specified burner turndown. The combustion control shall release from the low-fire position and adjust the firing rate in accordance with the load demand except that release may be delayed for a period not exceeding 60 seconds to insure stabilization of the flame. On cold startup, release may be delayed on units equipped with controls which prevent movement to high fire until a predetermined water temperature is reached. Unless a medium is specified (see 6.2), combustion controls shall be electrically, pneumatically, or hydraulically operated. Air for pneumatic combustion controls shall be contractor furnished or Government furnished, as specified (see 6.2). When combustion controls fluids are contractor furnished, the contractor shall be responsible for the quality of the fluids. When air supply is Government furnished, the manufacturer of the generator shall specify the required volume, pressure, and air quality for dryness, filtration, and being oil-free (see 6.4). Electrical combustion controls shall not exceed either a two-wire 120V, 60Hz system with separate equipment ground conductor or a two-wire nominal 120V, 60Hz system obtained by using an isolation transformer with secondary grounded. A means shall be provided to permit manual control of the firing rate. However, the system shall be so designed that it shall be impossible to put fuel control on manual while the air control remains on automatic or vice versa. The systems shall function to cause a normal shut-down in response to a predetermined pressure for a steam generator or water temperature for a hot water generator below the steam or relief valve setting when the load demand drops below the minimum firing rate provided by the control system, and results in a buildup of pressure or temperature to the predetermined set point. The combustion control system shall be one of the types as specified (see 1.2.3, 3.11.6.1, 3.11.6.2, 6.1(d), and 6.2). A schematic logic diagram of the combustion control system shall be furnished (see 6.3). Other system requirements are as follows:

(a) Means shall be provided to assure adequate primary air for the required fuel input to preclude the possibility of an unsafe fuel/air ratio condition.
(b) The coal feed flow may be taken as the rate of feed.
(c) The air/fuel mixture shall be maintained within safe limits as established by operation under any generator output conditions within the controllable operating range of the system.

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(d) Means shall be provided for setting minimum and maximum limits on fuel and air control systems to prevent these systems from providing fuel and air flows beyond the stable flame limits of the fuel burning system. These minimum and maximum limits shall be defined by the manufacturer and verified by operating tests (see 4.5.2).
(e) When changing the rate of furnace input, the air flow and fuel flow shall be changed simultaneously at the proper rates to maintain safe air/fuel ratio during and after the changes.
(f) Maintain furnace balance draft (zero draft) at the desired set point in the combustion chamber.
(g) Means shall be provided to prevent the control system from demanding a fuel-rich mixture. Control action toward more fuel and less air shall be blocked when air/fuel ratio decreases below preset value.
(h) Means shall be provided to cut back fuel to available air. If the air/fuel ratio should decrease below an appropriate preset value, control action toward less fuel and more air shall be taken.
(i) The combustion control system shall be interlocked with the coal fuel drive to balance the ash discharge with the firing rate.
(j) Means shall be provided to permit as much on-line maintenance of combustion control equipment as possible.
(k) Means shall be provided for calibrations and check testing of combustion control and associated safeguard equipment.
(l) Oxygen and combustible meter shall be provided as an operating guide when specified herein.

3.11.6.1 Parallel positioning control. Parallel positioning control with air/fuel ratio control shall include a master proportional controller which shall be connected to the main fluid header to sense the changes in load. The output of the controller shall transmit a signal simultaneously to the forced-draft damper and fuel feed or valve operators. A separate ratio-adjustment controller which has been calibrated for the system to prevent a fuel-rich mixture shall be utilized in the line to the fuel feed or valve and shall vary the signal for the fuel feed or flow proportionally to the air flow. This action shall change the fuel feed or flow to establish the optimum air/fuel ratio. A separate ratio-adjustment controller shall be used for each fuel. The parallel positioning control shall be capable of offsetting variations in pressure-drop characteristics in final control elements, fuel viscosity, air temperature, or barometric pressure. When switching between automatic and manual operation, the transfer shall be smooth. All controller adjustment and calibrations of fuel feed and air flow shall be made at the front of the control cabinet or panel with flush-mounted indicators provided to show adjustments and the results.

3.11.6.2 Parallel positioning with oxygen compensation control. Oxygen compensation shall be provided in addition to the parallel positioning control specified in 3.11.6.1. The oxygen analyzer shall send an impulse to the air/fuel ratio controller which shall automatically readjust excess air demand set point with boiler or hot-water generator load. The controlling system shall have the capability to bias the excess air/boiler load curve. The air/fuel ratio controller shall not accept signals outside the preset limits. Compensation shall be made in the steady-state excess air/boiler load curve for the variance in composition of the fuel to be burned.

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3.12 Fuel systems. Generators shall be furnished with factory installed fuel system as specified in 1.2.3. Fuel systems shall be equipped with the master fuel trips required in 3.11.1(q). Fuel systems shall have a combustion process turndown (minimum continuous safe firing rate), as specified (see 6.1(e), 6.2, and 6.5), but shall be not more than 33.3 percent of the maximum continuous firing rate.

3.12.1 Fluidized bed coal burner. The fluidized bed coal burner shall include a coal hopper-feeder, limestone hopper-feeder, fuel/air chamber, fluidized bed burner, air nozzles, ducts and piping with drive and damper, valves and actuators for the combustion process turndown (see 3.12). The burner shall be properly sized and interconnected to satisfy the functional requirements and not interfere with the combustion process. The fuel burning system shall provide means for safe start-up, operation and shut-down of the combustion process. This shall include appropriate openings and configurations in the component assemblies to permit suitable observation, measurement, and control of the combustion process. The fuel burning system design shall be based on the following requirements:

(a) The air supply equipment for combustion shall be properly sized, arranged, and supervised to insure a continuous, steady air flow adequate for all operating conditions.
(b) The air inlets, duct work, shall minimize contamination of the air supply by such materials as flue gas, water, and fuel.
(c) The air equipment shall supply proper air flow during any furnace pressure pulsations.
(d) Means shall be provided to control the feeder outlet temperature within limits suitable for coal and limestone being fired.
(e) Piping which connects the feeder systems to fluidized bed shall be kept to a minimum length, sized and arranged to minimize unequal distribution of the coal and limestone.
(f) Fuel (coal and limestone) shall be supplied to the furnace continuously and within stable flame limits, independent of the auxiliary ignition subsystems over the operating range of the fuel burning equipment.
(g) The limits of stable flame for each burner subsystem producing a separate flame envelope shall be determined in section 4.5.2.
(h) Means shall be provided for adequate observation of the conditions at each burner subsystem and its ignition zone space shall be provided for flame safeguard control equipment.
(i) Means shall be provided for rodding or cleaning of the coal and limestone feeders if coking of the coal should occur.
(j) Convenient access for maintenance shall be provided. Requirements of good housekeeping shall be recognized.

3.12.1.1 Coal hopper-feeder. The coal hopper shall be properly sized and arranged to insure a continuous, steady flow adequate for all operating requirements without bridging of the fuel. The hopper shall be constructed of steel plate having minimum thickness of 3/16-inch, and shall conform to ASTM A588. The hopper shall include a variable speed feeder to assure a turndown as specified herein for introduction of the coal into the combustion chamber. The feeder shall be designed with sufficient capacity range to allow for variations in size, quality, and moisture content of the coal as specified herein to hold a minimum one hour firing at rated capacity. Each feeder drive shall be protected with a shear pin and means provided to sound an alarm and light a red indicating light when material feed becomes inoperative while feed motor is running. All belt driven drives shall have safety guards that

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conform to ANSI B15.1 and adjustable motor mounts for belt tension adjustment and belt replacement. V-belts shall conform to ZZ-B-225. Sheaves shall conform to NEMA MG-1 for minimum electric motor sheave diameter for a given hp motor. Motor sheaves shall be of the adjustable pitch type which provides a 20 percent speed variation, selected so that the specified capacity will be delivered with sheaves at approximately mid-position. V-belt drives shall be delivered with sheaves at approximately mid-position. V-belt drives shall be selected for continuous duty in accordance with Engineering Standards of Rubber Manufacturer's Association with a service factor of 1.5. Bearings shall be of the type normally furnished by the generator manufacturer.

3.12.1.2 Limestone hopper-feeder. The limestone hopper shall be properly sized and arranged to insure a continuous steady flow adequate for all operating requirements without bridging of the crushed stone. The hopper shall include a variable speed feeder in accordance with 3.12.1.1 to adjust the volume of limestone being introduced to compensate for the varying sulphur contents of the coal being fired. The capacity of the hopper shall be sufficient to sustain the coal firing of 3.12.1.1.

3.12.2 Fluidized oil bed system. The fluidized oil bed system shall have burners of the oil/combustion air premix, multiport nozzle type. The burner nozzles shall be fabricated from materials sufficient to withstand the temperatures they are exposed to. Fluidized oil bed system shall be equipped with the safety interlocks, limit switches, and control valves specified in table II and III as shown in Figure 1. In addition, the systems shall include a fuel oil pump, an oil strainer, relief valve(s), pressure gage with pressure gage cocks, a pressure regulating valve and oil preheating equipment. Oil burners shall produce a clean, stable, and efficient flame at all firing rates. The burner flame shape shall fit the furnace without impingement (see 6.4), on or overheating of the furnace walls, and without extending to the furnace roof. Oil burners of the return flow type shall be equipped with a return flow safety shutoff and a spring-loaded check valve to prevent return oil leakage on safety shutdown. The fuel system may operate on either the constant pressure or the constant differential principle provided positive control and adequate atomizing pressure is maintained at all firing rates. Fuel oil piping shall conform to ANSI B31.1.

3.12.2.1 Fuel oil pump. The fuel oil pump shall be a motor-driven positive displacement or variable displacement metering type having sufficient capacity to deliver 125 percent of the volume of the specified grade of fuel oil required at maximum rated burner capacity at 125 percent of the designed burner pressure measured at the burner tip when handling oil at the specified temperature from a suction lift of not less than 15 feet.

3.12.2.2 Strainers. Duplex strainers, self-cleaning strainers, or a pair of simplex strainers having isolating shut off valves shall be installed at the fuel pump suction connection, and at such other locations as may be necessary to insure the delivery of clean oil to the burner regardless of any internal strainer.

3.12.2.3 Relief valves. Automatic oil pressure relief valves shall be provided on the discharge side of the fuel oil pump, the discharge side of the fuel oil heaters, and such other locations as may be necessary to prevent excess pressure to the burner. Relief valves shall be complete with return piping to the fuel oil suction or return line.

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3.12.2.4 Preheaters. When specified (see 6.2), preheating equipment shall be furnished in accordance with the following:

(a) Electric preheaters shall be furnished on units firing No. 4 or No. 5 (light) fuel oil if required for proper atomization.
(b) Electric preheaters shall be furnished on units firing No. 6, NSFO and when required, No. 5 (heavy) fuel oils for initial light-off of cold units. Preheaters shall have sufficient capacity for the firing rate occurring when the combustion controls are in the position for low-fire starts. Electric preheaters may be furnished in the same shell as the steam or water preheaters.
(c) Steam or hot water heaters, as specified (see 6.2), shall be furnished on units firing No. 6 and Navy special fuel oil, and when required by the type burners used, on No. 5 (heavy) fuel oil.

The preheating equipment shall handle the applicable grade of fuel oil delivered to the preheater inlet connection at the temperature specified (see 6.2). When preheated oil is used, fuel oil systems shall be equipped with a temperature gage.

3.12.2.4.1 Electric preheaters. Heaters rated at least 5,000 watts, and a nominal 240 volts (V), shall be controlled by a magnetic contactor having a magnetic coil operated by a thermostat. Heaters of smaller size rated at a nominal 240V, and less than 15A per circuit, may be controlled by a line voltage-thermostat switch or by a magnetic contactor with a coil operated by a thermostat. The electric power circuits shall also contain approved means for disconnection, and shall be protected by circuit breakers. Separate circuits shall be provided for limit switches and control circuits shall not exceed 120V, shall be two-wire with one side grounded, and shall be protected by circuit breakers. When adequate heavy oil temperature is reached and maintained by the steam or hot water heater, the thermostatic control or limit switch shall cut off the electric heater unless the heater is used to minimize the temperature fluctuations of the oil leaving the steam or water preheaters, or to provide additional heating for hot water heaters to maintain proper atomization temperatures.

3.12.2.4.2 Steam or hot water heaters. The steam or water preheater shall be designed in accordance with section VIII of the ASME Code. Oil pressure drop through the heater shall not exceed 15 psi. The heater shall be designed for boiler working pressures of steam and liquids at the corresponding steam or water temperature. The heater may be of the bare tube or extended surface type. The heater shall be equipped with an adjustable oil temperature regulator or aquastat and, when necessary, a steam pressure regulator, which will maintain the oil temperature within 5°F of the set temperature. The fuel oil pipes and valves shall be arranged to automatically recirculate hot oil through standby periods between firing cycles. The heating systems shall include provisions for detecting or preventing oil leakage into the boiler. On steam heaters, means for visually detecting oil in the heater condensate line shall be provided. For hot water heaters, an auxiliary heating medium shall be interposed between the oil and boiler water. Alternate means for preventing entrance of oil into the boiler will be acceptable, subject to approval by the contracting officer.

3.12.3 Fluidized gas bed system. Fluidized gas bed system shall have burners of the gas/combustion premix, multiport spud type. The burner spuds shall be fabricated from materials sufficient to withstand the temperatures

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they are exposed to. Gas fuel systems shall be equipped with the safety interlocks, limit switches, and control valves specified in table II and III. The system shall include a drip leg and gas strainer installed ahead of the main pressure regulator. A pressure gage shall also be installed between the downstream safety-shutoff valve and the lubricated plug valve. The burner shall incorporate an effective means of preventing flame propagation back into the mixing chamber, or gas or air supply. The combustion air and gas supply shall be controlled to prevent gas from entering the burner until primary air is available, and to shut off the gas in case of primary air failure. When provided with the specified type of gas at the specified pressure, gas burners shall produce a stable flame which stays in one position. There shall be no blowoff from the burner bed and no pulsating noise or vibration of the flame. Burner parts shall be protected against overheating. Fuel ports shall be machined or otherwise accurately made. The burner flame shape shall fit the furnace without impingement (see 6.4) on, or overheating of, furnace walls and without extending to the furnace roof. Gas piping shall conform to the requirements of ANSI Z223.1, and ANSI Z83.3. Otherwise gas burners shall conform to ANSI Z83.3 for systems with intermittent igniter as shown in Figure 1.

3.13 Ignition systems. Generators shall be equipped with ignition systems designed to impinge on the main air/fuel mixture, and supply sufficient ignition energy to provide positive ignition of all fuel discharged to the fluidized bed under interlock light-off conditions. The igniter shall be integrated with the programming control system specified herein. Ignition systems shall be equipped with safety interlocks, limit switches and control valves as shown in Figure 1 and specified in table III. All igniters shall be of the intermittent type with gas-electric ignition. Ignition transformers shall be rated on the secondary side at not less than 6,000V for gas-electric igniters. A normally open valve, UL listed for gas service, equipped for automatic closing and spring opening shall vent the igniter gas piping to the atmosphere (outside of the generator room) between the two safety-shutoff valves if the igniter is rated in excess of 120,000 Btu/hr. Igniter gas piping shall conform to ANSI Z83.3.

3.14 Control panel. The controls, including operating switches, indicating lights, gages, alarms, motor starters, fuses, and circuit elements of control systems shall be mounted on a single control panel or cabinet insofar as practicable in order to centralize the control functions, and shall be designed for separate mounting. Switches shall include burner start, manual-automatic combustion control, fuel oil heater, and fuel selector switches, as applicable. A schematic wiring diagram of the control circuits shall be furnished (see 6.3). Colored indicating lights shall be provided and shall be activated when the interlock for the completion of the operation is activated in control sequences herein. The arrangement and colors of the lights shall be as specified in table IV and the lights shall be permanently identified. The control panel or cabinet or the individual elements shall be provided with NEMA 12 dust protective covers, and shall be mounted on a solid base and not subject to vibrations or be a hazard for the operator. All terminals requiring connection upon installation of the generator shall be permanently identified in such a manner that correct connections can be easily made by reference to the instructions and wiring diagrams provided with the equipment. Terminal connections for control functions shall be segregated from power circuit terminals. Wiring shall be in accordance with NFPA No. 70.

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Unless otherwise specified (see 6.2), the control panel shall be mounted at the right side when viewed facing the burner end of the generator. When specified (see 6.2), a remote free-standing control panel shall be furnished and include construction details (see 6.4) covering cabinet, location, mounted controls, and instruments. Flame and igniter failure indicating lights shall remain mounted in a cabinet on the generator even when a remote free-standing control panel is used. A duplicate set of colored indicating lights shall be furnished on the remote panel. When specified (see 6.2), a main circuit breaker shall be furnished and mounted either adjacent to the control panel or packaged separately for remote panel mounting at the site, as specified (see 6.2). The following listed instruments and controls shall be flush mounted on the front of the panel but not limited to:

(a) Recording meters of 12 inch diameter, 24 hour per revolution circular chart type or 7 inch strip chart shall be provided to indicate, record, and integrate steam flow, and indicate and record air flow and flue gas temperatures. Recorders shall be equipped with an electric chart drive, condensate chambers, capillary tubing, orifice plate, and 100 charts with ink supply shall be provided for field installation.
(b) Steam pressure gage having a minimum diameter of 6 inches with a graduated range that will result in normal pressure indication within the middle range scale in addition to the one mounted on the boiler in 3.10.
(c) Pressure gages of a minimum diameter of 4-1/2 inches shall be provided to indicate gas and oil fuel supply pressure and gas and oil fuel pressure at burners shown in Figure 1.
(d) Draft gages of vertical scale type to indicate in inches of water column pressures at burners, furnace boiler exit, and induced draft fan discharge.
(e) Smoke density indicator with strip chart recorder, when specified in 3.14.2.

3.14.1 Annunciator. When specified (see 6.2), an annunciator shall be provided in accordance with ISA S18.1. The annunciator shall indicate all nonrecycling burner shutdowns specified in 3.11.3, include high and low feedwater level operational control alarms, provide at least two spare stations or points in addition to the others specified. The annunciator shall not require special or additional limit devices, or change sensitivity of limit switches, or annunciate without the limit device being actuated. The annunciator shall provide means to indicate the first-limit or off-limit condition causing burner shutdown and to sound an alarm. The annunciator shall also have a test button and a feature capability to store information on additional causes of burner shutdown. Subsequent causes of burner shutdown will be indicated after the operator acknowledges the first burner shutdown. A dust-tight enclosure shall be provided. The annunciator shall include additional features and information required in section 5, Specifications, of ISA S18.1, as specified (see 6.2). The following alarms shall indicate off-standard conditions to alarm in advance of safety shutdown but not limited to the following:

(a) Low fuel oil supply pressure.
(b) Low atomizing steam or air oil pressure.
(c) High furnace pressure.
(d) High furnace draft.

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(e) Low combustion air flow.
(f) Loss of interlock energy.
(g) Loss of combustion control energy.
(h) Proof of fuel valve closed at start of cycle.
(i) Low bed temperature.
(j) Smoke density reading, when specified herein.

3.14.2 Smoke density indicator and transmitter. When specified (see 6.1(j) and 6.2), an electronic opacity monitoring device with strip chart recorder and transmitter for stack mounting shall be provided to detect, indicate, alarm, and record visible emissions from the stack. The system shall conform to the requirements of CFR Title 40, Part 60.344, Method 9, "Visible Emissions." A scale indicator shall be provided to display the output of the smoke density transmitter calibrated in Ringlemann. The instrument need not have clean air self-cleaning feature. The instrument shall operate on normal 120V (minimum 102V, maximum 132V), 60 Hz, and have internal electronic voltage regulation with power consumption of 20A maximum.

3.15 Draft fans. The draft fans shall be for a balanced draft system; shall be integrally or separately mounted as specified (see 6.2); and shall be variable speed or fixed speed electric-motor-driven for direct or belt drive, as specified (see 6.2). The centrifugal or axial draft fans shall be rated for pressure, power, flow rate, speed of rotation, and efficiency in accordance with the test methods of AMCA 210D. The fan shall be of simple, rugged construction, designed and balanced for minimum power consumption, noise, and vibration, and shall be designed and constructed to permit easy access for adjustment, maintenance, and repair. A manual control switch shall be provided to permit operation of the fan for purging the furnace prior to a cold start. Fan noise shall not exceed 90 decibels (dB) at 3-foot stations at a height of 4.5 feet when enclosed in the generator outer casing. The fan noise shall drop at least 6 dB when measured at a distance of 6 feet and 4.5 feet in height. All exposed belt driven fans shall have safety guards that conform to ANSI B15.1 and adjustable motor mounts for belt tension adjustment and belt replacement. V-belts shall conform to ZZ-B-225. Sheaves shall conform to NEMA MG-1 for minimum electric motor sheave diameter for a given hp motor. Motor sheaves shall be of the adjustable pitch type which provides a 20 percent speed variation, selected so that the specified capacity will be delivered with sheaves at approximately mid-position. V-belt drives shall be selected for continuous duty in accordance with Engineering Standards of Rubber Manufacturer's Association with a service factor of 1.5. Bearings for integral mounted fans shall be of the type normally furnished by the generator manufacturer. Bearings for separately mounted fans shall be precision antifriction bearings that meet the requirements of FF-B-171 for an average life of 100,000 hours; the self-alining sheave type; or roller bearings mounted in suitable pillow blocks as specified (see 6.2). Oil reservoirs shall be equipped with an oil level sight gage and drain plug. The draft fans shall be sized for an output volume of combustion air or hot combustion flue gases sufficient for complete combustion at maximum rated burner output plus an additional 10 percent. Complete combustion shall be based on the percent of excess air necessary to obtain at least the minimum carbon dioxide flue gas content listed in table I. The balanced draft system shall have a forced draft fan and an induced draft fan, designed as follows:

3.15.1 Forced draft fan(s). The static pressure at the fan(s) outlet plus the maximum burner and fluidized bed pressure drop shall give a balanced draft required in 3.11.6(f) within the combustion chamber. Centrifugal fans shall have backward curved blades or radial blades.

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3.15.2 Induced draft fan(s). The outlet static pressure required at the induced draft fan(s) shall be as specified (see 6.2). The induced draft fan(s) shall be designed for handling hot combustion flue gases at a maximum temperature, as specified (see 6.2), and shall have water-cooled bearings when operating over 600°F to maintain a balanced draft required in 3.11.6(f) within the combustion chamber. Centrifugal fans for induced draft service shall have radial-tip or forward-blade designs.

3.16 Electrical wiring and equipment.

3.16.1 Electric motors. Electric motors for the driven auxiliaries shall be constructed and rated in accordance with the applicable provisions of NEMA MG-1. Motors shall be general purpose, continuous duty, totally enclosed fan-cooled, ball-bearing, induction motors. The motors shall be designed for operation on a power supply having the specified characteristics (see 6.2).

3.16.2 Motor starters. Unless otherwise specified (see 6.2), motors shall be provided with magnetic across-the-line starters with overload and under-voltage protection and manual-off-automatic control stations. Magnetic starters shall conform to the applicable requirements of NEMA ICS-1, ICS-2, and ICS-6 and shall have NEMA type 12 enclosures.

3.16.3 Wiring. All wiring shall conform to and be installed in accordance within the applicable provisions of NFPA No. 70. Circuits for the boiler operating and safety controls shall either be a two-wire 120V 60Hz system with separate equipment ground conductor or a two-wire nominal 120V 60Hz system obtained by using an isolation transformer with secondary grounded. Limit controls shall be wired on the hot side of the control circuit. Further detailed requirements for the control circuitry on both high pressure boilers and low pressure units shall be as specified in section I and section IV of the ASME Boiler and Pressure Vessel Code. When a free standing control panel is required (see 3.14) burners shall be wired for remote starting. A shematic wiring diagram of the power circuits shall be furnished (see 6.3).

3.17 Stack. When specified (see 6.2), stack sections, stack supports, umbrella collar and cap, and flue transition piece (if required), shall be provided. Unless otherwise specified (see 6.2), the stack shall be 20 feet in height when assembled on the generator and measured from the ground line. Stack sections shall be sheet steel having a thickness of not less than 0.0972 inch. When required, flue transition pieces shall be furnished to connect from the flue weight of the stack. The transition piece shall be capable of carrying the full weight of the stack. A ladder or other means shall be provided to make access to the stack mounted smoke density transmitter when specified in 3.14.2. Such ladder or means shall conform to OSHA 1910.27.

3.18 Condensate return system. When specified (see 6.2), a condensate return system shall be furnished with each steam unit. The system shall conform to type I, style B of MIL-B-17749 and shall be sized to deliver not less than twice the total feedwater required at maximum rated output of the unit. The capacity of the condensate receiver shall be not less than one-fifth the volume of water converted to steam per hour.

3.19 Economizer. When specified (see 6.1(l) and 6.2), a separately packaged economizer shall be provided for the generator. The economizer shall be complete with prefabricated interconnecting breeching for attachment to the generator flue outlet and the feed piping between the economizer and the generator shell, and shall require minimum field assembly. Economizers shall

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be of the manufacturer's standard design for the operating conditions specified herein and be designed in accordance with ASME Boiler and Pressure Vessel Code Section I. Materials and components (valves, fittings, piping) used in construction shall withstand the temperatures and pressures prevailing under maximum load conditions. Flow and temperature controls for the economizer feedwater shall be furnished to preclude condensation of the flue gases in accordance with 3.8.6. Provisions shall be made in the valving to allow bypass of the economizer feedwater control to permit maintenance of the control valves. Soot blower(s) of the air-puff or steam type, as specified (see 6.2), shall be provided and shall be capable of cleaning all surfaces of soot deposits on economizers furnished. Air-puff soot-blowers shall be designed for 200 psig air. The 200 psig air source shall be furnished by the economizer manufacturer, others, or the Government, as specified (see 6.2).

3.20 Cleaning, treatment and painting. Unless otherwise specified (see 6.2), surfaces normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice as long as the process meets the criteria specified herein. The color of the finish coat except for surfaces to be painted aluminum shall be in accordance with FED-STD-595, number as specified (see 6.2). Component parts, such as motors, etc., procured factory finished need not be repainted. The paint shall be free from runs, sags, orange peel, or other defects.

3.20.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, mill scale, products of corrosion, dirt or other foreign substances. When aluminum heat-resisting paint is required (see 3.20.4), all exterior surfaces of the generator shall be cleaned by commercial blast before application of the insulation. All exterior and interior surfaces of the sheet steel jacket shall be cleaned by commercial brush-off blast prior to assembly. Mechanical cleaning need not be performed on sheet steel jacket with straw colored mill scale, if the surfaces are free of all loose mill scale and products of corrosion.

3.20.2 Treatment. All metal surfaces, except for units to be painted with aluminum heat-resisting paint, shall be treated to increase adherence of the coating system. A primer pretreatment coating shall be applied after cleaning and before any evidence of rust or other contamination can result.

3.20.3 Manufacturers' standard painting. The manufactuers' standard materials and standard processes shall be used to apply paint consisting of not less than one coat of rust-inhibiting primer and one coat of finish enamel with a total dry film thickness of not less than 2.5 mils. If the paint is electrostatically applied, it shall consist of not less than one coat of finish enamel with an average of not less than 2.5 mils.

3.20.4 Aluminum heat-resisting painting. When specified (see 6.2), before assembly, the generator shell and the interior of the jacket shall be finished with one coat of aluminum heat-resisting paint conforming to TT-P-28. The paint shall be applied directly to clean bare metal surfaces and shall attain a minimum dry film thickness of 1 mil. After assembly, all exposed surfaces of the generator and associated equipment normally painted in commercial practice shall be cleaned as specified herein and finished with two coats of the aluminum heat-resisting paint. Each coat shall be applied to a minimum dry film thickness of 1 mil.

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3.21 Technical manuals. The contractor shall provide technical manual(s) conforming to the Data Item Description(s) (DID) listed on the Contract Data Requirements List (CDRL) (DD Form 1423) (see 6.3).

3.22 Instructions. Each unit shall be furnished with operating instructions (see 6.3) prepared in a permanent, legible form suitable for posting on or near the unit. The operating instructions shall include information relating to the normal ignition and shut down of the unit and shall otherwise comply with the recommendations for operating instructions as given in UL726 or UL795, or both, as applicable. The following caution sign shall be prominently located on all hot water generators in 1/4 inch letters:

CAUTION

RAPID REPLACEMENT OF YOUR HOT WATER WITH RETURN WATER AT TEMPERATURE BELOW THE BOILER MANUFACTURER'S RECOMMENDATIONS CAN CAUSE SERIOUS DAMAGE TO YOUR BOILER. DO NOT ALLOW THIS TO HAPPEN. DO NOT FIRE YOUR BURNER UNTIL AFTER WATER FLOW HAS BEEN ESTABLISHED IN ALL ZONES AND THROUGH BOILER.

3.23 Marking. Each unit shall be marked with the manufacturer's name, catalog designation, electrical rating, type of flooring approved for mounting, installation clearances, and other information required under UL726 or UL795, as applicable.

3.23.1 Military marking. When specified (see 6.2), the contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data conforming to MIL-STD-130 in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly and/or subassembly and parts as directed by the Government inspector. Brass screws, rivets, or bolts of not less than 1/8-inch in diameter shall be used to affix the plates.

3.24 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be treated for moisture and fungus resistance in accordance with MIL-T-13867 as specified herein. Electrical components with a temperature rise of 40° Centigrade (C) or less shall be coated in accordance with type II of MIL-T-13867 with fungus resistant varnish conforming to MIL-V-173. Components with a temperature rise exceeding 40°C shall be coated in accordance with type III or IV of MIL-T-13867 with varnish conforming to MIL-I-24092 with grade and class of varnish dependent on the component, class of insulation above 40°C and the recommendations of the manufacturer of the equipment.

3.25 Electromagnetic interference suppression. When specified (see 6.2), the unit shall meet the electromagnetic interference suppression requirements and test limits for class I and class IIB equipment as specified in MIL-STD-461.

3.26 Workmanship. The quality of workmanship shall be such as to produce generators that meet the requirements of this specification and standards prevailing among manufacturers normally producing this equipment.

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4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.

(a) First article (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) On-site inspection (see 4.2.3).

4.2.1 First article inspection. First article inspection shall be performed on one generator when a first article is required (see 3.2 and 6.5). This inspection shall include the examination of 4.3 and the tests of 4.4.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on each unit being produced for or offered to the Government. This inspection shall consist of the examination of 4.3 and the tests of 4.4.1 and 4.4.5. Presence of one or more defects or failure of any test shall be cause for rejection of the unit.

4.2.3 On-site inspection. Testing shall be performed on all generators at the site after installation. This testing shall be in addition to quality conformance inspection performed at the factory and shall be supplemental to first article inspection, if such inspection is specified. On-site testing shall consist of tests of 4.4.2 and when specified herein 4.4.4, and all other tests deemed necessary by the procuring activity when specified in the contract to verify compliance with the requirements of this specification. On-site testing shall be performed either by the generator manufacturer or by the installing contractor, as specified (see 6.2). The manufacturer shall have the privilege of representation at tests performed by others. When the manufacturer is responsible for on-site tests, the detailed requirements and schedule for the test program will be as specified in the contract (see 6.2). In all cases, deficiencies revealed by on-site testing shall be corrected at the generator manufacturer's expense and any required retesting shall also be at the manufacturer's expense.

4.3 Examination. Each boiler shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.3.1 Standards compliance. The manufacturer shall make available to the contracting officer or his authorized representative evidence of compliance with the applicable standard(s) cited in 3.5. Certification specified in 3.5.1 will be accepted as evidence of compliance.

4.3.2 Design verification. Unless otherwise specified (see 6.2 and 6.3), the manufacturer shall furnish drawings and computations necessary to verify compliance with the heat release criteria of 3.8.4, and efficiencies of 3.8.3.

4.4 Tests.

(a) Hydrostatic tests (see 4.4.1).
(b) On-site operational tests (see 4.4.2).
(c) Fuel-to-steam and fuel-to-hot water efficiency tests (see 4.4.3).
(d) Tests for particulate, nitrogen, and sulfur emissions (see 4.4.4).
(e) Factory auxiliary equipment tests (see 4.4.5).
(f) Electromagnetic interference tests (see 4.4.6).

The hydrostatic and operational tests shall be performed on each generator supplied under a contract, including the first article sample, if furnished. Efficiency tests of 4.4.3 shall be performed only on the first article sample. The electromagnetic interference tests will be performed on one unit when electromagnetic interference suppression is supplied.

4.4.1 Hydrostatic tests.

4.4.1.1 Watersides. Waterside shall be hydrostatically tested in accordance with the requirements of the ASME Boiler and Pressure Vessel Code. The appropriate certification specified in 3.5.1 will be accepted as evidence of compliance.

4.4.1.2 Fuel systems. The part of the preassembled fuel system furnished integrally with the boiler shall be checked for leakage during the fire testing of the boiler. The soap bubble or the halogen-sniff test method shall be used to verify tightness of the gas system.

4.4.2 On-site operational tests. Each generator shall be fire-tested to demonstrate control and operational conformance and combustion efficiency to requirements of 3.8.3.1, 3.11.6(d), and 3.12.1(g) and to the requirements of this specification under varying load conditions ranging from the specified capacity to the minimum burner turndown ratio without on-off cycling. All burners shall be tested for ability to start a cold generator in accordance with the manufacturer's instructions. Boilers that exhibit (1) excessive or unexplained loss of ignition, (2) nuisance shutdown due to faulty burner or control operation, (3) improper flame, (4) excessive carbon deposits, or (5) necessity for frequent or difficult adjustments shall be rejected. Control and operational test shall be as specified in 4.4.2.1 through 4.4.2.7.

4.4.2.1 Sequencing. The generator shall start, operate, and stop in strict accordance with the specified operating sequence.

4.4.2.2 Flame failure. The operation of the flame failure control shall be verified by simulated flame and ignition failures. The igniter establishing period, the main-burner establishing period, flame-failure reaction time, and the safety-shutoff valve closing times shall be verified by stop watch (see 3.11.2 and table IV).

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4.4.2.2.1 Immunity to hot refractory. The generator shall be operated at high fire until the combustion chamber reaches maximum temperature (1600°F minimum). Each main fuel supply and igniter valve shall then be closed manually and the generator allowed to cool. After the thermocouples cool to the minimum set temperature and the alarm sounds, the combustion safeguard shall drop out causing the safety shutoff valves to close within the specified flame-failure reaction time.

4.4.2.2.2 Minimum ignitor test. The fuel supply to the igniter flame shall be gradually reduced to the point where the combustion safeguard begins to drop out (sense "no flame") but holds in until the main fuel valve opens. At this point of reduced igniter fuel supply, the igniter flame shall be capable of safely igniting the main burner. If the main fuel valve or supply can be opened on an igniter flame of insufficient intensity to safely light the main flame, the boiler shall be rejected.

4.4.2.3 Spark rejection test. With the fuel supply to both the igniter and main flame closed, the generator shall be operated into the ignition cycle. When the spark ignition comes on, the flame detector shall not be activated. If the flame detector can be activated, the generator shall be rejected.

4.4.2.4 Burner turndown. The specified burner turndown for the modulating control shall be verified by firing at the minimum firing rate required by the specified burner turndown (see 3.12).

4.4.2.5 Boiler limit and fuel safety interlock. Safety shutdown shall be caused by simulating interlock actuating conditions for each boiler limit and fuel safety interlock. Safety shutdowns shall occur in the specified manner.

4.4.2.6 Combustion controls. The accuracy range and smoothness of operation of the combustion controls shall be demonstrated by varying the load demand through the entire firing range required by the specified burner turndown. The control accuracy shall be as specified in 3.8.5.

4.4.2.7 Safety valves. The high-pressure limit switch shall be locked out or otherwise made inoperative and the generator safety valves shall be lifted by steam. The relieving capacity, popping pressure, blowdown, and reseating pressure shall be determined by observation and measurement to be in accordance with the ASME Boiler and Pressure Vessel Code. This test shall be made in the presence of the contracting officer or his authorized representative. The ASME standard symbol will be accepted only as indicating compliance with the design and material requirements of the code.

4.4.2.8 Blowdown valves and try cocks. All blowdown valves and try cocks shall be tested for proper operation.

4.4.3 Fuel-to-steam and fuel-to-hot water efficiency tests. The fuel-to-steam and fuel-to-hot water efficiency, and steam quality at the specified capacity of the generator shall be determined in accordance with the ABMA Standard Test Procedure for Packaged Firetube Boilers or ASME PTC 4.1. The efficiency shall be determined by the Indirect Method. Performance shall be as specified in 3.8.3.2 and 3.8.3.3. Test runs shall be made at the maximum rated capacity.

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4.4.4 Tests for particulate and nitrogen emissions. When specified (see 6.1(k), and 6.2), tests for emissions of particulates, nitrogen, and sulfur dioxide shall be conducted on-site of installation and the method of test for particulate emissions shall be as specified (see 6.2). Tests shall be performed by the generator manufacturer, or the installing contractor, as specified (see 6.2). The manufacturer shall have the privilege of representation at tests performed by others. The location of tests shall be as specified (see 6.2).

4.4.5 Factory auxiliary equipment test. Draft fans, fuel oil heaters, fuel pumps, dampers, actuators, valves, and electric motors shall be operational tested to determine compliance with the referenced specifications or standards. Standard approval symbols or certification of compliance (see 6.3) from certifying organizations may be accepted at the discretion of the contracting officer or his authorized representative. The operation of fans, fuel oil heaters, fuel pumps, and electric motors shall be closely observed during the on-site operational tests (see 4.5.2) for possible defects or nonconformance.

4.4.6 Electromagnetic interference suppression tests. To determine conformance to 3.24, the first article shall be subjected to the tests in accordance with MIL-STD-462. When suppressed to conform to 3.24, the manufacturer upon approval of the contracting officer may furnish a certification (see 6.3) in lieu of the test that the generator meets the requirements, together with a list of the suppression devices installed. The list shall be sufficiently detailed to allow visual determination that the devices are installed.

4.5 Packaging inspection. The inspection of the preservation, packaging, packing, and marking shall be in accordance with the requirements of section 4 of MIL-B-3180. The inspection shall consist of the quality conformance inspection of MIL-B-3180. When specified (see 6.2), a preproduction pack shall be furnished for examination and test and within the time frame required in accordance with MIL-B-3180.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), the generators shall be preserved, packaged, packed, and marked in accordance with MIL-B-3180. The level of preservation and packaging and level of packing shall be as specified (see 6.2).

6. NOTES

6.1 Intended use. The generators covered by this specification are intended for generation of steam to be used in processing operation, space heating, or power generating systems. Hot water generators are for space heating and may also be used for service water. The generators as specified with mechanical collectors alone will probably not meet existing particulate emission standards, without additional emission control equipment. The boiler type (coal with oil or gas or a combination of the three types) should be determined in accordance with the availability and economics of the fuel supply. The following factors should be noted:

(a) Supervised manual start up is the available method since the generator has multiple burners. When multiple burners exist adjacent burners must not be detected by the flame sensor.

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(b) The boiler or high temperature water generator should be sized to permit efficient burner operation under low load conditions during the summer.
(c) Repetitive self-checking circuits have been included as standard because generators operating in occupied buildings or in hazardous locations require such safety controls.
(d) The type of combustion control system should be selected in accordance with generator input capacity of 1.2.1 and load changes expected.
(e) Burner turndowns higher than those required or useful should not be specified. Generators are usually limited to a minimum firing rate of 33.3 percent of maximum firing rate with an option minimum firing rate of 16.7 percent of the maximum firing rate with oxygen compensation.
(f) Temperature-anticipating master combustion controllers should be specified for generators intended for use with high temperature water generators when the generator capacity exceeds 30,000,000 Btu/h.
(g) Normally, oil-fired generators above 5,000,000 Btu/h output should be capable of burning fuel oil grades No. 2, No. 4, No. 5, and No. 6. When changing grade of fuel, replacement may be necessary (1) of the fuel nozzles because of the different fuel viscosity, and (2) of the fuel pump motor to drive the pump at a speed that ensures required fuel pressure in order to maintain the required burner output and meet the optimum air/fuel ratios.
(h) Gas-electric igniters as specified herein are required for UL, FM and Industrial Risk Insurers requirements. Oil igniters are not available.
(i) Standard design pressures for steam generators covered by this specification range from 30 psi to 300 psi and for hot water generators to 160 psi.
(j) A smoke density indicator should be specified on all generators over 5,000,000 Btu/h input capacity.
(k) If the generator is subject to environment regulation, each agency should consult with their environment support office since regulation limits of particulate and emissions vary from city, state, and Federal governments.
(l) Economizers can increase fuel to steam efficiencies 1 to 1-1/2 percent for 150 psig generators and 2-1/2 to 3 percent for 250 psig generators.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, date, and number of this specification.
(b) Type, size, and capacity of boiler required; design pressure for types II and III; burner type and grade of fuel oil or type of gas to be burned; type of combustion control required (see 1.2.1, 1.2.2, 1.2.3, 3.8.2, and 3.11.4).
(c) When a first article is required (see 3.2, 4.2.1, and 6.5).
(d) If different supply the required fuel characteristics (proximate analysis, ultimate analysis, and heating value) of intended coal, limestone, oil, and gas fuel (see 3.8).
(e) Gas supply pressure at the site upstream of the pressure regulator (see 3.8).
(f) Operating pressure for steam generators; operating temperature and pressure for hot water generators; when water temperature is other than 200°F (see 3.8.1).

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(g) When the generators shall conform to the particulate and nitrogen, sulfur oxide emission tests and required particulate and nitrogen oxide emission limits (see 3.8.3.1 and 6.1(l)).
(h) When an explosion relief door or device is required (see 3.9.5).
(i) Whether air for carbon reinjection shall be furnished by others or the Government (see 3.9.8)
(j) Type of ash removal required (see 3.9.10).
(k) When the furnace is to be started on gas and oil (see 3.11).
(l) When indicator shall record bed temperature and type of chart required (see 3.11.5).
(m) Type of combustion control required and medium of operation required, electric, electronic, pneumatic, or hydraulic and who is to supply air for pneumatic controls (see 3.11.6 and 6.1(d)).
(n) When oxygen compensation is required (see 3.11.6.2 and 6.1(d)).
(o) Required burner turndown expresses as percentage of maximum continuous firing rate (see 3.12, 6.1(e), and 6.4).
(p) When preheaters are required with types and temperature of the oil at each preheater inlet (see 3.12.2.4).
(q) When mounting of control panel is different (see 3.14).
(r) When remote free-standing control panel is required and details of its construction, location, mounted controls, instruments, and wiring for remote starting (see 3.14 and 3.16.2).
(s) When a main circuit breaker is to be furnished and the location of mounting (see 3.14).
(t) When an annunciator is required, and additional features and information required in accordance with ISA S18.1 (see 3.14.1).
(u) When smoke density indicator is required (see 3.14.2 and 6.1(j)).
(v) Required type of mounting for draft fan; variable speed or fixed speed motors; direct or V-belt electric-motor-driven; and type of bearings required for separate mounted fans (see 3.15).
(w) Outlet static pressure required of the induced draft fan; and the maximum temperature of the flue gases (see 3.15.1 and 3.15.2).
(x) Characteristics of the electrical power supply available at the site (see 3.16.1).
(y) Type of motor starter, if different (see 3.16.2).
(z) When stack is required and design details of installation (see 3.17).
(aa) When a condensate return system is required (see 3.18).
(bb) When economizer is required and type of economizer, and who is to supply air (see 3.19 and 6.1(o)).
(cc) When painting is to be other than manufacturer's standard; color finish coat required (see 3.20).
(dd) When surfaces are to be painted with aluminum heat-resisting paint conforming to TT-P-28 (see 3.20.4).
(ee) When identification marking in accordance with MIL-STD-130 is required (see 3.23.1).
(ff) When treatment for fungus resistance is required (see 3.24).
(gg) When electromagnetic interference suppression is required (see 3.25 and 4.4.6).
(hh) Whether on-site inspection will be performed by the generator manufacturer, or by the installing contractor; outline of tests if generator manufacturer will perform the tests (see 4.2.3).
(ii) When design verification is not required (see 4.3.2).
(jj) When generator shall be tested for emissions of particulates and oxides of sulfur and nitrogen; method of testing, who is to perform tests, location of tests (see 4.4.4).

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(kk) When a preproduction pack is required and time frame required for submission of the preproduction pack (see 4.5).
(ll) Level of preservation and packaging and level of packing required (see 5.1).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENTS	APPLICABLE DD 1664
3.5.1, 4.4.5, 4.4.6	Certification of compliance	DI-E-2121
3.11.1, 3.11.6, 3.14, 3.16.3, 4.3.2	Installation drawings	DI-E-7031
3.9.8, 3.11.6, 4.3.2	Design data	DI-R-24040A
3.21, 3.22	Publication, commercial	DI-M-24010A
3.5.1, 3.8.7	Reports, tests	DI-T-2072

(Copies of Data Item Descriptions required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

6.4 Definitions. The following definitions are applicable for use by the contractor and acquisition agency in the interpretation and implementation of the requirements of this specification. Other terms used in this specification should be interpreted in accordance with the definitions of UL726, UL795, NFPA No. 85B, 85D, 85E, ABMA LEXICON, ANSI Z83.3 or ASHRAE No. 85.

Burner turndown. The burner turndown shall be the minimum continuous safe firing rate which can be effected automatically by the combustion control system during the normal firing cycle without interruption or change of burner components. The burner turndown shall be expressed as a percentage of the maximum continuous firing rate.

Effective radiant heating surface. The ERHS shall be the heat exchange surface within the furnace limits which is directly exposed to the radiant heat of the flame on one side and the water being heated on the other. This surface shall be expressed in square feet and shall be equal to the product of the interior mean perimeter of the furnace and the furnace height as defined herein, exclusive of the fluidized bed.

Furnace volume. Furnace volume shall be expressed in cubic feet and shall be the product of the interior cross-sectional area and height of the furnace. The furnace shall be considered as not extending beyond the inlet of the firetubes in the convection section of the generator.

Flame impingement. Flame impingement means physical contact of the burner flame on the surface (fireside) carrying the products of combustion of the furnace or any tube and tube sheet of the boiler so that the flame path is diverted at these surfaces.

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Ignition temperature. Ignition temperature means the lowest temperature at which a gas-air mixture may ignite and continue to burn. This is also referred to as the auto-ignition temperature. For safety with manufactured gases a temperature of about 1200°F is needed and for natural gas a temperature of about 1400°F is needed (see NFPA No 86A).

Nonproprietary controls. Controls marketed by a supplier on a nonexclusive right to sell.

Nonrecycling safety-shutdown. Stopping burner operation by shutting off all fuel and ignition energy to the furnace by means of a safety interlock or safety controls and requiring a manual reset.

Recycling shutdown. Stopping of a burner operation by a normal shutdown when certain interlocks and limit switches have sensed a limit or off-limit condition and then allows the burner to automatically recycle when the condition corrects itself.

Supervised manual system. A supervised manual system is one in which a well-trained operator has primary responsibility for the safe start-up, operation and normal shutdown of a generator with interlocks to insure that the operation follows established safe procedures. This system includes certain interlocks for checking operator action, certain safety trips and flame supervisions as outlined in 3.11.1. The status of start-up sequence should be indicated to the operator.

Total generator heating surface. The total generator heating surface shall be expressed in square feet and shall consist of those surfaces which are exposed to the products of combustion on one side and water on the other. The total generator heating surface shall be measured on the side receiving the heat and shall include the radiant heating surface defined herein plus the firetubes and the portions of such other convection surfaces which are below the normal water line.

6.5 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete fluidized bed boiler unit or it may be a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangement for examinations, tests, and approval of the first article.

Custodians:

Army - ME
Navy - YD
Air Force - 99

Preparing activity:

Navy - YD

Project No. 4410-0052

Review activities:

Army - CE
Veterans Administration - DA

User activities:

Navy - CG, MC

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TABLE I. Requirements for combustion.

Requirements	Coal 11,500 Btu/lb Percent			Natural Gas 1,000 Btu/ SCF Percent			Light Oils No. 2, JP-5, Diesel Percent			Medium Oils No. 4, NSFO Percent			Heavy Oils No. 5, & 6 Percent		
Average Combustion Efficiency	81			81			81			81			81		
Rated load	40	70	100	40	70	100	40	70	100	40	70	100	40	70	100
Carbon dioxide (minimum)	10.0	12.6	13.6	8.5	9.0	10.0	10.5	11.5	12.5	11.0	12.0	13.0	12.0	12.6	13.6
Oxygen (minimum)	5.0	4.0	3.0	6.0	5.0	4.0	6.0	5.0	4.0	6.0	5.0	4.0	6.0	5.0	4.0
Carbon monoxide (maximum)	0.2	0.2	0.2	*	*	*	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.05
Excess air (maximum)	40	25	20	40	25	20	40	30	20	40	30	20	40	30	20
Equivalent Bacharach Smoke Scale (maximum)	2	2	2	0	0	0	2	2	2	3	3	3	3	3	3

*Units rated below 25,000,000 Btu/h input capacity shall meet UL 795 requirements. Units rated above 25,000,000 Btu/h input capacity shall meet 0.05 percent maximum.

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TABLE II. Requirements for safety interlock and limit switches.

Safety interlock and limit switch	Type of burner shutdown required Boiler size No. 1 to 4	 Boiler size All others
a. Each steam generator to be equipped with two low water level cutout switches, to be either float actuated or of the electrode type in accordance with 3.10.1. Each hot water generator to be equipped with one low water level cutoff installed as recommended in section IV of the ASME Boiler and Pressure Vessel Code and low water shall activiate a separate red indicating light.	Nonrecycling (see 6.4)	Nonrecycling
b. High steam pressure or high water temperature limit switch, each to be supplemental to and separate from the operating control.	May recycle (see 6.4)	May recycle
c. Air-flow switch of the pressure type to monitor forced air from purge and firing cycles; to be furnished on all sizes of units, whether or not unit is equipped with centrifugal switch. (Fan motor interlock) See Figure 1.	Nonrecycling	Nonrecycling
d. Air-flow switch of the pressure differential type to monitor induced air-flow for purge and firing cycles to be furnished on all sizes of units, whether or not unit is equipped with centrifugal switch. (Fan motor interlock) See Figure 1.	Nonrecycling	Nonrecycling
e. Low fire start interlock to prove damper and fuel flow control for coal drive, oil and gas valves are in low-fire position for light-off. Interlock to be part of or integrated with combustion control system.	Stop cycle until low-fire position is proven.	Stop cycle until low-fire-position is proven.

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TABLE II. Requirements for safety interlock and limit switches. (continued)

Safety interlock and limit switch	Type of burner shutdown required Boiler size No. 1 to 4	 Boiler size All others
f. Low voltage interlock if not furnished as an inherent characteristic of the flame failure system.	May recycle	Nonrecycling
g. Fuel system interlocks specified in table III.	As required in table III	As required in table III
h. Low water flow interlock on high temperature water generators rated above 15 psi or 250°F whichever is greater. Low water flow shall activate the same separate red indicating light as in "a" above.	Nonrecycling	Nonrecycling

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TABLE III. Requirements for gas ignition system and gas and oil fuel system components.

Item (see Figure 1)	Description
	REQUIREMENTS FOR IGNITION SYSTEM FOR COAL, GAS, AND OIL FIRING
A	Atmosphere vent valve. Normally open, UL listed for gas service equipped for automatic closing and spring opening, if total input is in excess of 120,000 Btu/h.
B	Igniter gas pressure regulator suitable for maximum specified inlet pressure (see 3.8).
C	Pressure gage conforming to GG-G-76.
D	Manual lubricated plug valves. Two required, one located at inlet to gas train, the other downstream of both safety shutoff valves to permit leakage testing of valves with leakage test connections.
E	Igniter safety shutoff valve. Two required, to be solenoid-operated. Valves shall be spring closing and rated to close at a pressure higher than line pressure upstream from the pressure regulator and shall be 5 psi minimum. Valve closing time shall not exceed one second.
F	Low gas supply pressure trip, switch to be manual reset type, to cause a nonrecycling shutdown.
G	High gas supply pressure trip, switch to be manual reset type, to cause a nonrecycling shutdown required only when total input is in excess of 120,000 Btu/h.
H	Strainer.
LK	Leak test bleed valves, 3 required.
W	Manual supervisory and shutoff valve, approved for gas service, with double pole double through (DPDT) switch.
	REQUIREMENTS FOR MAIN GAS FUEL SYSTEM COMPONENTS.
J	Main gas safety shutoff valves. Two approved motorized safety shutoff valves, normally closed, electric reset and holding. Valves shall be spring closing and rated to close at a pressure higher than line pressure upstream of the regulator "P" and shall be 5 psi minimum (see 3.8). Valve closing time shall not exceed one second.

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TABLE III. Requirements for gas ignition system and gas and oil fuel system components. (continued).

Item (see Figure 1)	Description
K	Individual burner gas safety shutoff valves, approved motorized, normally closed electric reset and holding shall have overtravel interlock in closed position. Valves shall be spring closing and rated to close at a pressure higher than line pressure upstream of the regulator "P" and shall be 5 psi minimum (see 3.8). Valve closing time shall not exceed one second.
L	Atmosphere vent valve, manual sized to ASME CSD-1.
M	Atmosphere vent valve. General purpose, approved for gas service normally open, full ported, electrically-operated solenoid valve connected in a manner to vent piping between safety shutoff valves whenever these valves are closed.
N	Main gas control valve, approved motorized operational valve to control fuel flow rate for modulating combustion control systems. Two in series, one operates on air/ratio and one operates on temperature.
P	Constant gas line pressure regulator approved suitable for maximum specified inlet pressure (see 3.8).
Q	Cleaner or strainer with drip leg ahead.
R	High gas supply pressure trip switch to be of manual reset type, to cause a nonrecycle shutdown.
S	Low gas supply pressure trip switch to be of manual reset type, to cause a nonrecycle shutdown.
T	Pressure gage conforming to GG-G-76.
V	Manual shutoff.
W	Manual supervisory and shutoff valve, approved for gas service with (DPDT) switch.
X	Nonelectric valve held open by a fusible link designed to close the valve at 165°F shall be installed at the furnace in the main fuel line.
LK	Leak test bleed valves, 5 required.

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TABLE III. Requirements for gas ignition system and gas and oil fuel system components. (continued).

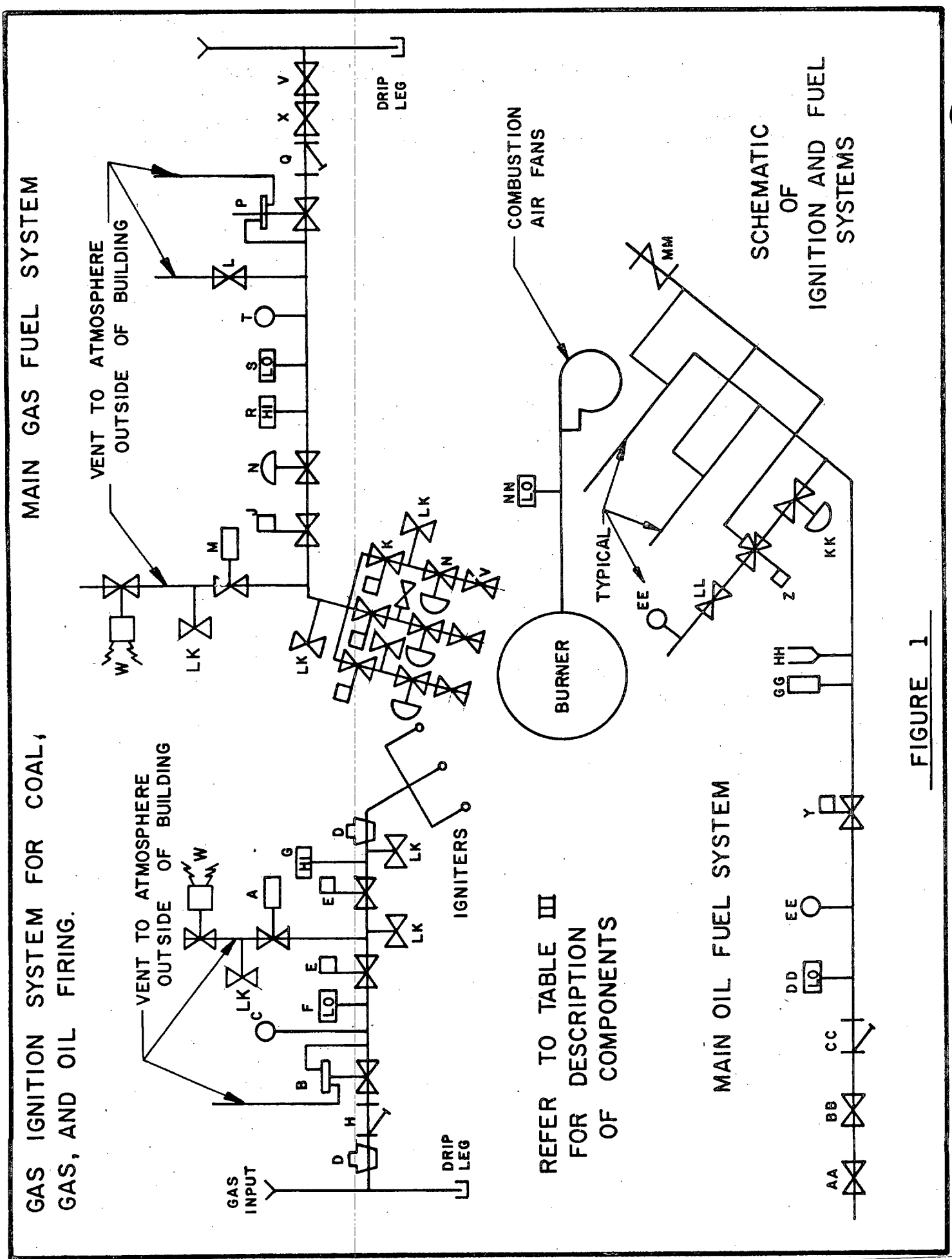
Item (see Figure 1)	Description
	REQUIREMENTS FOR MAIN OIL FUEL SYSTEM COMPONENTS.
Y	Main oil safety shutoff valve, solenoid operated, normally closed, valve to be electric reset and holding, valve to be spring closing, shall be UL listed or FM approved for oil service, and shall have a valve closing time not to exceed one second.
Z	Individual burner safety shutoff threeway valve and shall have overtravel interlock in closed position otherwise same as valve "Y".
AA	Manual shutoff valve.
BB	Nonelectric valve held open by a fusible link designed to close the valve at 165°F, shall be installed at the furnace in the main fuel line.
CC	Strainer.
DD	Low fuel pressure switch to be of manual reset type, to cause a nonrecycle shutdown.
EE	Individual burner pressure gage conforming to GG-G-76.
FF	Main line pressure gage conforming to GG-G-76.
GG	Thermometer.
HH	Low- and excessive-oil temperature interlock switches for units with oil preheaters, may recycle after sounding an alarm. Installed in a different tapping than the operating control.
KK	Individual oil control valve, approved motorized operational valve to control fuel flow rate for modulating combustion control system.
LL	Individual burner supervisory and shutoff valve.
MM	Check valve.
NN	Low combustion air interlock switches, one for each forced fan upstream, one for each induced fan downstream to cause a nonrecycling shutdown (see Table II).

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TABLE IV. Arrangement and colors of indicating lights.

Operation	Color
Low water	Red
Safety lockout	Red
Atomization on, medium, when used	Yellow
Purge complete, where required	White
Ignition on	Amber
Main fuel on	Green
Power on, where required	Blue

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-B-29205 BOILERS, FLUIDIZED BED, STEAM AND HOT WATER, HIGH AND LOW PRESSURE, FIRETUBE, PACKAGED TYPE (2,000,000 TO 50,000,000 BTU/H

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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